



University of Balamand
Faculty of Arts & Sciences
CSIS 200 Introduction to Computers & Programming

2-D Arrays

1. Write a main program that declares a 5x5 integer array called **M**, stores in the array **M** the result of the multiplication of 1->5 by 1->5.

2. Write a main program that declares a 5x5 character array called **shapes**, stores in the array the following stars:

Shapes	0	1	2	3	4
0	*	*	*	*	*
1	*	*			
2	*		*		
3	*			*	
4	*				*

3. Write a main program that declares and initializes a 3x4 integer array of values from your choice, calls method **sum** that takes a 2-D array, computes and returns the sum of all elements in the array.

4. Write a main program that declares and initializes a 3x5 double array called **grades** of values from your choice, finds and displays the lowest and largest element in the array.

5. Write a main program that declares and initializes a two dimensional array 3x4 called **A** to values from your choice. Call method called copy that takes array **A**, copy the elements of the array **A** into a one dimensional array **B**. Your method should return array **B**.

6. Write a main program that:

- Declares and initializes a 3x4 double array called M of values from your choice.
- Declares a one-dimensional array N.
- Stores the largest element of each row of array M in the array N.

7. Suppose the weekly hours for all employees are stored in a two-dimensional array. Each row records an employee's five day working hours with five columns. For example, the array shown below stores the work hours for 5 employees.

	M	T	W	H	F
Emp. 1	5	8	8	8	9
Emp. 2	8	8	8	8	8
Emp. 3	8	9	8	7	9
Emp. 4	9	6	4	8	8
Emp. 5	7	7	7	8	9

Write a main program that asks the user to enter the daily working hours for each employee, computes and displays the total working hours for each employee per week if it is greater than or equal to 40. Otherwise, display "You still need to work more **x** hours" (where x represents the number of hours the employee has to work to reach 40 hours).